Sophia Xing

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Education

Rice University Expected May 2027

B.A. in Computer Science, B.A. in Mathematics, and Minor in Business *Houston, Texas*

- **Relevant Coursework:** Algorithms, Computer Systems, Concurrent Programming, Programming Languages, and Statistics
- **Leadership:** Computer Science Club President, Rice D2K Lab Datathon Director, Chinese Student Association Vice President, Women in Computing Vice President, Undergraduate Investment Fund Senior Analyst, Women in STEM Mentor, Club Tennis

Professional Experience

Google Aug 2026 – Present

Software Engineer Intern, Finance Engineering *San Jose, California*

- Optimized **concurrent Python retrieval**, parallelizing documentation and issue searches to reduce median LLM calls by **40%**.
- Redesigned **multi-agent** troubleshooting with intent-based routing and prioritized retrieval across documentation and issue trackers.
- Engineered fault-tolerant async execution, isolating tool failures to preserve concurrent retrieval workflows.
- Integrated **backend RPC services** into AI agents, implementing authenticated data retrieval and structured financial workflows.

NVIDIA May 2026 – Aug 2026

Software Engineer Intern, GeForce NOW *Santa Clara, California*

- Engineered **distributed storage orchestration** for GPU sessions, supporting persistent and ephemeral volumes across lifecycles.
- Built resource provisioning and validation in Java and Spring Boot, coordinating allocation and assignment across services.
- Prevented duplicate session operations using **Redis locks** and **idempotent state transitions** across concurrent requests.
- Improved **scheduler fault tolerance** by handling recoverable allocation failures and preserving sessions for retry.
- Integrated Google Ads into session queueing, implementing routing and auth to support **$5M+ in projected annual revenue**.

Amazon Web Services (AWS) May 2025 – Aug 2025

Software Engineer Intern, EC2 *Seattle, Washington*

- Enhanced the **Elastic Load Balancer Console** for 1M+ customers by building an Infrastructure-as-Code aware interface.
- Designed a tag-based system to detect and map **Kubernetes** resources, improving cross-service navigation and visibility.
- Implemented IaC specific filters and indicators that cut resource discovery time by **30%** using React Query caching.
- Developed a service layer integrating the **EKS API** via AWS SDK v3, optimizing data fetching and reducing API calls.

Episode Six (Payment Technologies) Jun 2024 – Aug 2024

Software Engineer Intern *Austin, Texas*

- Optimized Java backend APIs for authentication and transactions, improving security and reducing support time by **40%**.
- Engineered validation pipelines to prevent incorrect payments and ensure reconciliation accuracy across merchant networks.
- Integrated **3D Secure (3DS) OTP authentication**, strengthening transaction security and compliance with global standards.

Yale University Nov 2022 – Aug 2023

AI Research Assistant, Medical Data Retrieval *Remote*

- Built a **GPT-3** medical query engine using **approximate query processing**, reducing latency from seconds to under **100 ms**.
- Developed a Transformer pipeline translating SQL queries into natural language, improving usability for nontechnical users.
- Fine-tuned an **LLM** on 10K+ hybrid medical queries, improving response accuracy and contextual relevance by **25%**.

Projects

Data Pipeline Orchestrator | Two Sigma-Sponsored Capstone | *C#, .NET, Azure Functions, Azure Service Bus, PostgreSQL*

- Built a **DAG parser** validating cycles, dependencies, data bindings, and compute requirements for distributed pipelines.
- Implemented a **stateless, event-driven scheduler** with atomic dependency tracking for concurrent fan-out and fan-in execution.
- Engineered fault-tolerant execution with retry mechanisms and persistent state recovery across distributed tasks.

OwlDB | *Go, REST APIs, Concurrent Programming*

- Engineered an **in-memory NoSQL database** supporting hierarchical documents, CRUD requests, and JSON Schema validation.
- Implemented a **concurrent skip-list index** with fine-grained locking, atomic updates, and efficient ordered range queries.
- Built token-based authentication and **real-time SSE subscriptions** with event broadcasting and concurrent client management.

SpectralWalk | *C++, Python, Fast Fourier Transform (FFT), Spectral Graph Theory*

- Building a **spectral analysis engine** using FFT to compute eigenvalues of modular random walks in **O(n log n)** time.
- Developing an **optimization solver** to identify fastest-mixing transition probabilities and generating sets on cyclic graphs.

Technical Skills

Languages: Python, Java, C++, C#, Go, JavaScript, TypeScript, SQL, C, Bash

Frameworks & Technologies: Spring Boot, React, Node.js, .NET, AWS, Azure, Docker, Kubernetes, PostgreSQL, Redis, Git, Linux